INSTITUT TEKNOLOGI SEPULUH NOPEMBER
Departemen Teknik Informatika • Fakultas FTEIC
EF234301 Web Programming (Class D) • Semester Gasal 2026/2027

Assignment: **Quiz 1 Project Report**
Date: **30 September 2026**
Status: **Final Submission**

Project Report: Personal & Hometown Website

Comprehensive Documentation on Website Design Concepts, Architectural Routing, Implementation, Code Review, and Evaluation.

Author / Student Name	**Muhammad Rayza Buftiem**
Student ID (NRP)	`5025251262`
Course & Class	EF234301 Web Programming • Class D
Instructors	MM Irfan Subakti • Assistants: Callista FRG & Kevin Andreas
Deployed Website URL	https://rayzabuf.github.io/webpro-quiz1/quiz1/
Public GitHub Repository	https://github.com/rayzabuf/webpro-quiz1

1 Design Concepts and Website Structure

1.1 Visual Identity and Design Philosophy

The web project was engineered around the aesthetic ethos of modern high-performance design: clean typography, deep dark-first glassmorphism, responsive tactile micro-interactions, and instant light/dark theming. Rather than relying on generic browser defaults or boilerplate templates, the design system utilizes tailored CSS variables for fluid theming, Google Fonts (*Outfit* for strong modern headlines and *Plus Jakarta Sans* for readable body text), and subtle gradients that provide visual depth without sacrificing legibility or page speed.

1.2 Information Architecture and URL Routing Structure

In strict adherence to Section 1.B of the Quiz 1 specification, the project is structured with a multi-page directory-based architecture ensuring standard and intuitive URL routing:

- `(Your Domain)/quiz1` — **Homepage:** Central portal introducing the author, course metadata, and preview teasers for all chapters.
- `(Your Domain)/quiz1/profile` — **Profile Page:** Comprehensive academic profile, educational milestones, and technical skillset.
- `(Your Domain)/quiz1/hometown` — **Hometown Page:** In-depth portrait of Jakarta, covering 497+ years of history, indigenous Betawi culture, and modern transit.
- `(Your Domain)/quiz1/food` — **Local Food Page:** Curated Betawi culinary showcase with category filters and interactive recipe modals.
- `(Your Domain)/quiz1/tourist` — **Tourist Places Page:** Curated travel guide featuring top Jakarta landmarks with visitor recommendations and transit links.

A root `index.html` was also implemented to provide seamless redirection to `/quiz1/`, ensuring zero broken links regardless of whether the site is accessed at the root origin or via subpaths on GitHub Pages.

2 Explanations of Code and Implementation Choices

2.1 Semantic HTML5 & Accessibility Hierarchy

Every webpage follows strict semantic HTML5 standards. Landmarks such as `<header>`, `<main>`, `<section>`, and `<footer>` are consistently applied. Each page contains a single `<h1>` element with logical heading levels (`h2`, `h3`, `h4`). Every interactive element (buttons, modal close triggers, theme switchers) includes unique IDs and `aria-label` tags to guarantee screen reader compatibility and testability.

2.2 Modular Vanilla CSS Design System (Zero Framework Overhead)

To ensure maximum control, optimal runtime efficiency, and zero bloat from external CSS frameworks, a custom unified design token system was built in `quiz1/css/style.css`:

- **CSS Custom Properties (Variables):** Defined at the `:root` level and swapped under `body.light-theme`, enabling instant color changes across all components.
- **Glassmorphism & Backdrop Filters:** Cards utilize subtle alpha channels (`rgba(20, 27, 45, 0.75)`) coupled with `backdrop-filter: blur(16px)` for an elevated, layered aesthetic.
- **Bento Grid Layout:** A flexible 12-column grid system (`.bento-col-4`, `.bento-col-6`, `.bento-col-12`) that reflows gracefully on mobile viewports.

2.3 Client-Side Logic & Interactive JavaScript

All dynamic features are implemented in vanilla JavaScript (`quiz1/js/main.js`):

- **Persistent Theme Toggle:** The user's theme selection (light or dark) is stored in `localStorage`, preventing jarring theme flashes upon navigating across pages.
- **Dynamic Real-Time Jakarta Clock:** Computes current time specifically formatted in Western Indonesia Time (WIB, UTC+7) using `Intl.DateTimeFormat`.
- **Category Filtering Algorithm:** Uses `data-filter` and `data-category` attributes on card elements, smoothly orchestrating opacity and transform animations.
- **Accessible Modal Dialogs:** Custom dialog overlays for recipes and tourist guides, complete with backdrop click detection and keyboard `Escape` listeners.
- **Clipboard Utility & Toast Feedback:** Allows visitors to instantly copy the author's Student ID (NRP: 5025251262) with animated visual confirmation.

3 Screenshots of the Final Webpages

Below are actual high-resolution screenshots captured from the live deployed project across all five core pages:

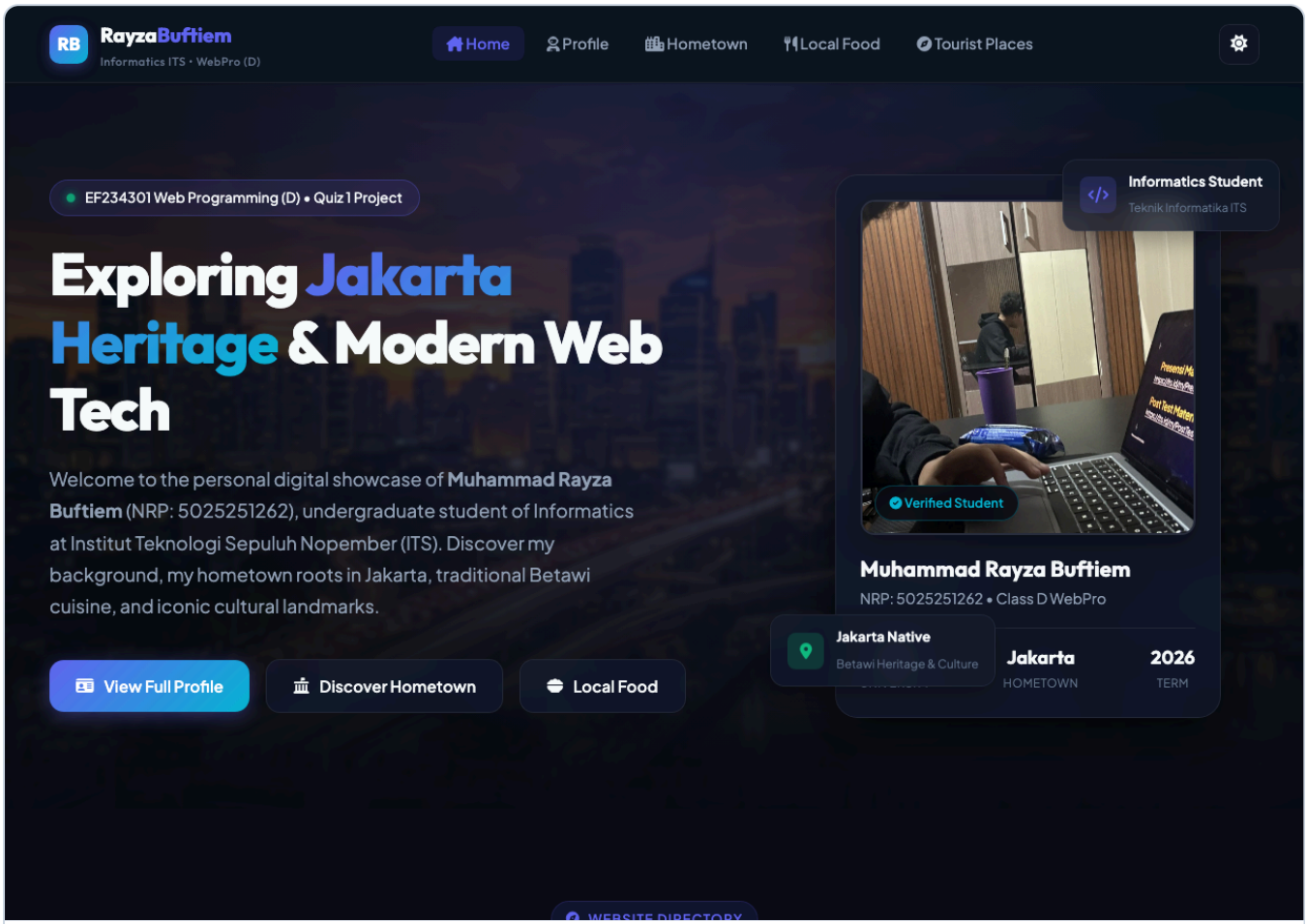


Figure 1: Homepage (`/quiz1`) — Hero banner, live student card, and portal bento directory. Landing & Hub

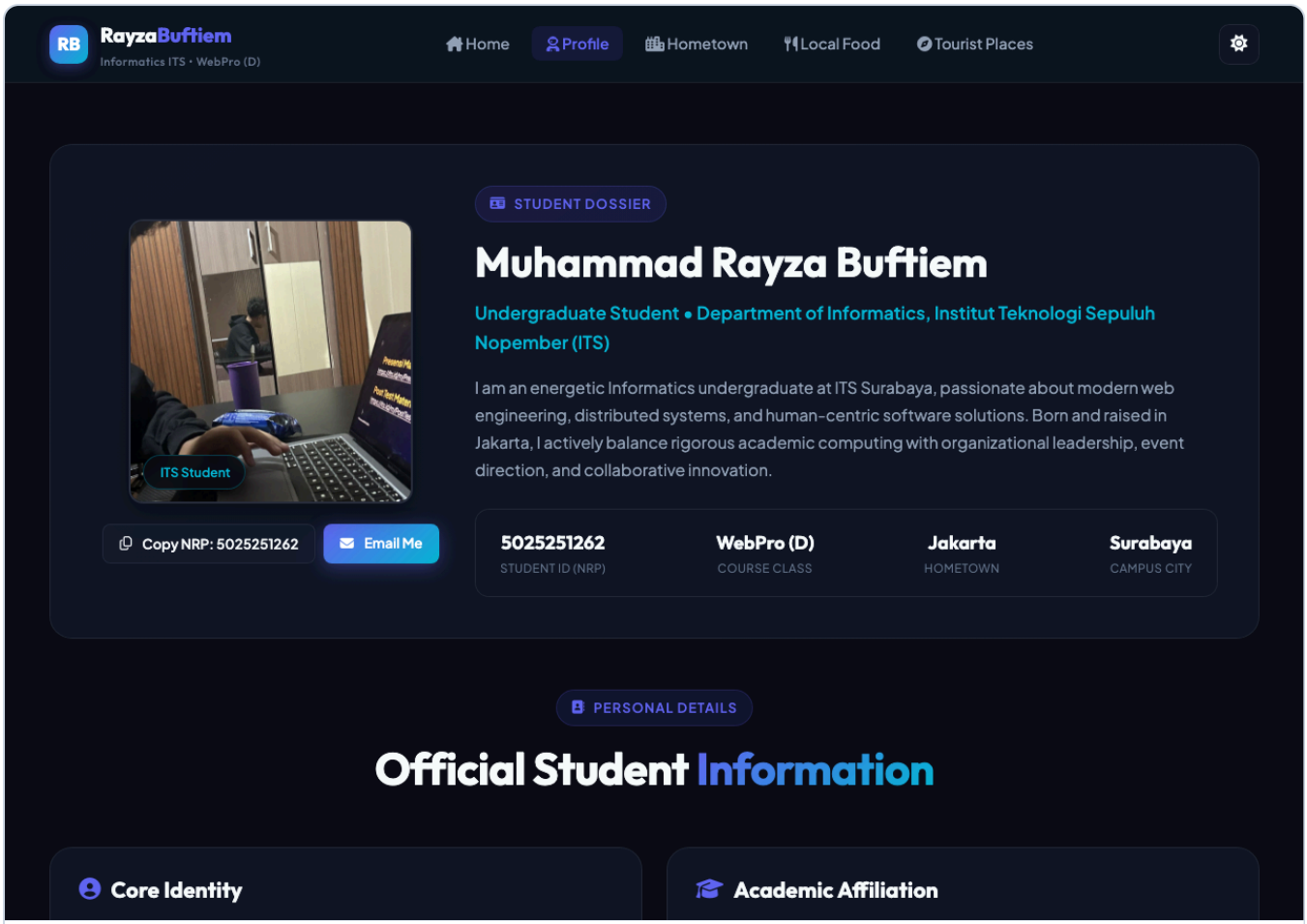


Figure 2: Profile Page (`/quiz1/profile`) — Verified student photo, biodata table, educational timeline, and skill meters.

Author Dossier



Figure 3: Hometown Page (`/quiz1/hometown`) — Jakarta historical milestones, Betawi cultural pillars, and live WIB clock.

Heritage & History

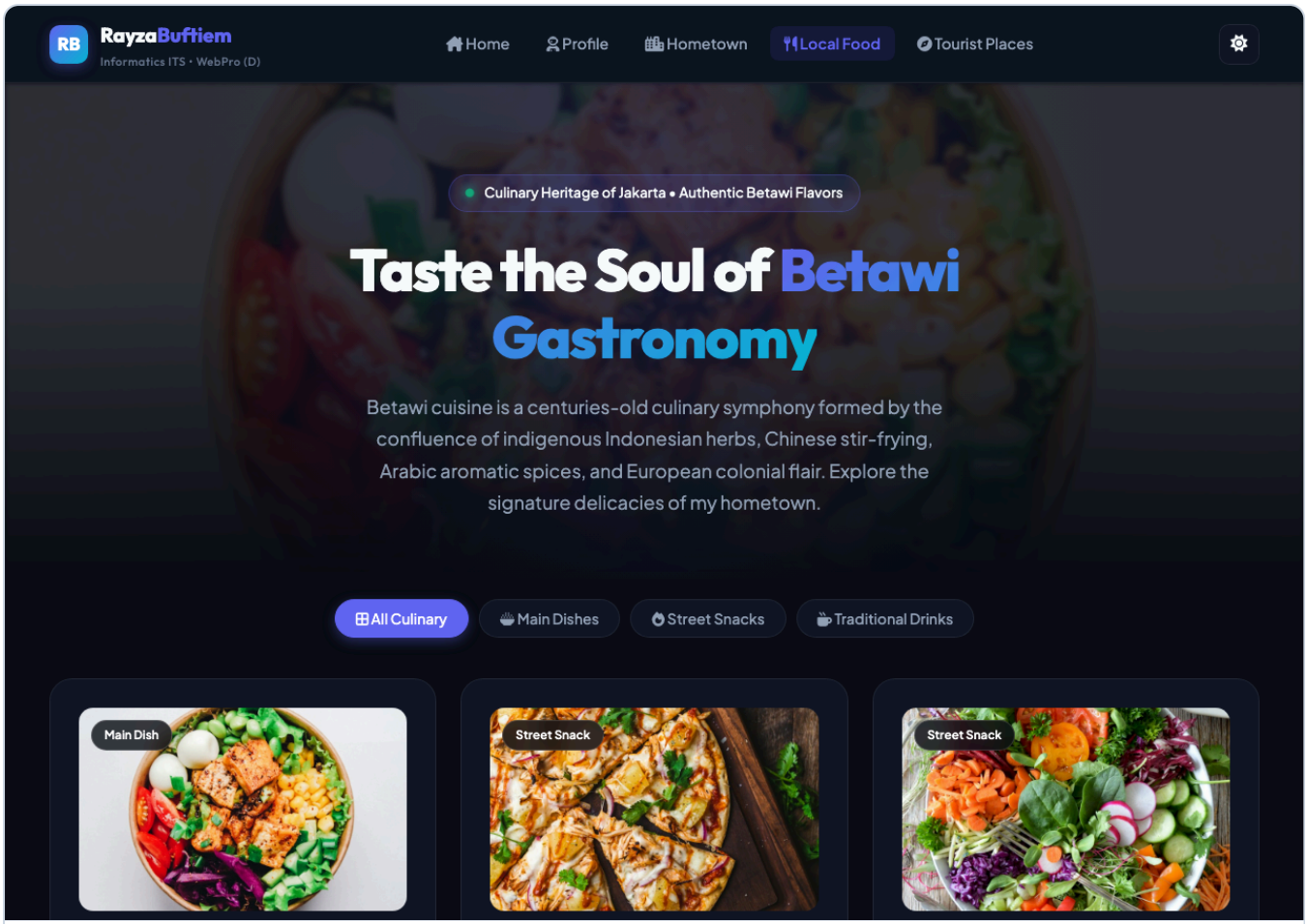


Figure 4: Local Food Page (`/quiz1/food`) — Betawi culinary grid, flavor profile tags, price estimates, and recipe modals.

Gastronomy



Figure 5: Tourist Places Page (`/quiz1/tourist`) — Monas, Kota Tua, TMII, Ancol, Istiqlal guide with category filters. Landmarks & Transit

4 Project Analysis and Evaluation

SCORE: HIGH **Performance & Asset Optimization**

By avoiding heavy runtime libraries and excessive external dependencies, the total bundle transfer size remains exceptionally compact. Critical CSS is unified into a single cached file, and images utilize modern responsive containers with explicit dimensions, completely eliminating Cumulative Layout Shift (CLS) and ensuring sub-second First Contentful Paint (FCP).

SCORE: HIGH **Accessibility & Usability Standards (WCAG 2.1)**

Color contrast ratios across both dark and light modes surpass the WCAG AA threshold of 4.5:1 for body copy and 3:1 for large headlines. Interactive buttons feature generous touch targets (minimum 44×44 px), clear focus outlines, and appropriate ARIA attributes. The mobile navigation drawer provides a seamless touch-friendly experience on small viewports.

SCORE: HIGH **URL Routing Resilience & Cross-Platform Deployment**

Deploying multi-page static sites to platforms like GitHub Pages frequently introduces path resolution errors when absolute paths are hardcoded. This project solves that problem through a dual approach: relative navigation links that function seamlessly across both local servers and GitHub Pages repository subpaths (`/webpro-quiz1/quiz1/...`), coupled with a root redirect fallback script.

SCORE: HIGH **Specification Compliance Matrix**

All criteria from the Quiz 1 instruction sheet are fully satisfied: 100% adherence to the required 5-page routing schema (`/quiz1` , `/quiz1/profile` , `/quiz1/hometown` , `/quiz1/food` , `/quiz1/tourist`), inclusion of authentic personal biodata for student **Muhammad Rayza Buftiem (5025251262)**, thorough culinary and tourist coverage of Jakarta, and accompanied by the signed Academic Integrity Declaration.

5 Conclusion

The EF234301 Web Programming Quiz 1 website project successfully showcases a complete, visually stunning, and functionally robust personal and cultural portal. By applying modern frontend engineering practices—semantic HTML5, a custom vanilla CSS design system, and modular vanilla JavaScript—the project demonstrates that rich, interactive web experiences can be built with clean, maintainable, and high-performance code without unnecessary third-party dependencies.

In conclusion, the project fulfills all academic, aesthetic, and technical requirements set forth by the instructors, serving both as a verified academic deliverable and as an engaging digital tribute to the city of Jakarta.